LINUX

The Anime Drama Beginner's Guide

Learn Linux like a story: simple explanations, character chats, terminal commands, mini missions and interview practice.

Featuring: Kai the Linux Mentor • Miko the Curious Beginner • Captain Root

ZERO PANIC. ONE COMMAND AT A TIME.

A friendly study book for absolute beginners and aspiring DevOps / Cloud Engineers.

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How to use this book

Imagine Linux is a huge kingdom. Your computer is the kingdom, files are its rooms, users are its citizens, and the terminal is the control room. You do not need to memorize everything today. Understand the idea, try the command, then explain it in your own words.

> Miko:
> “Kai, the black screen looks scary. What if I break the computer?”

> Kai:
> “A terminal is just a way to give instructions. Start with safe commands that only look around. Read before you change anything.”

Your learning route

Chapter	What you will learn
1. Linux basics	Linux, distributions, kernel, shell and terminal
2. Files & folders	Filesystem tree, paths, navigation and file commands
3. Permissions	Users, groups, chmod, chown, umask, sudo and sticky bit
4. Processes & services	PID, signals, jobs, systemd, logs and cron
5. Pipes & troubleshooting	Redirection, grep, sort, wc, tail and log analysis
6. Networking basics	IP, ports, DNS, ping, curl, ss and troubleshooting
7. DevOps practice	A mini incident, revision sheet and interview answers

Study trick: Say the command out loud, type it, observe the result, and explain what happened. That is how you turn theory into confidence.

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1. Welcome to the Linux kingdom

Linux is an operating system family. It helps your computer manage hardware, memory, files, users, programs and networks. Servers often use Linux because it is stable, flexible, secure when configured well, and easy to automate.

Miko:
“So Linux is like the manager of a hotel?”

Kai:
“Exactly! It assigns rooms, manages staff, checks who can enter, and keeps the hotel running.”

Important words

Word	Child-friendly meaning
Kernel	The heart of the operating system. It talks to hardware and manages resources.
Shell	A command interpreter. It reads your typed instructions.
Terminal	The window where you type commands and see results.
Distribution (distro)	A packaged Linux operating system, such as Ubuntu, Debian, or Amazon Linux.
Root user	The super-administrator with very powerful permissions.
CLI	Command Line Interface — controlling the computer by typing.
GUI	Graphical User Interface — controlling the computer with windows and buttons.

Linux boot story — what happens when you switch on?

Step	What happens
1. BIOS / UEFI	The machine checks hardware and starts the boot process.
2. Bootloader	A program such as GRUB helps load the Linux kernel.
3. Kernel	The kernel initializes hardware and core system functions.
4. init / systemd	The first userspace process starts services and the system.
5. Login	You reach a login screen or terminal and can start working.

Captain Root:
“Remember: kernel = heart, shell = translator, terminal = your microphone.”

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2. Explore the filesystem

Linux stores files in one big tree. The top of the tree is /, called root directory. Do not confuse / with /root: / is the top-level directory; /root is the home folder of the root user.

Miko:
“Where is my Desktop folder?”

Kai:
“First, ask where you are. Then look around. Linux navigation is like finding your way through a school.”

The important directories

Directory	What lives there?
/	The top of the entire filesystem tree.
/home	Personal folders for normal users.
/root	Home directory of the root administrator.
/etc	System configuration files.
/var	Frequently changing data, including many logs.
/var/log	System and application log files.
/tmp	Temporary files; contents may be cleaned automatically.
/bin and /usr/bin	Common user commands and executable programs.
/sbin and /usr/sbin	System administration commands (layout varies by distro).
/dev	Device files representing hardware and virtual devices.
/proc	Virtual filesystem with process and kernel information.
/opt	Optional/add-on application software.
/mnt and /media	Common mount points for filesystems and removable media.

Navigation commands

```
pwd                 # show the current directory\nls                  # list files\nls -l          # de
```

Absolute vs relative path

Path type	Example	Meaning
Absolute	/home/miko/notes.txt	Full path starting from /.
Relative	Documents/notes.txt	Path from your current directory.
Special	.. or .	Parent directory / current directory.

Create, copy, move and remove

```
mkdir practice              # create a directory\ntouch notes.txt              # create an empty file\ncp not
```


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Safety: Be very careful with `rm -r` and especially `rm -rf`. They can delete many files without asking. Never run a command you do not understand on a production server.

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3. Users, groups and permissions

Linux is a shared kingdom. Different people can use the same server, but not everyone should open every file. Permissions decide who can read, edit or run a file.

Miko:
“Why can't every user edit every file? That would be easier!”

Kai:
“And dangerous! Imagine every student could edit the school's exam results. Permissions protect important files.”

Three permission types

Symbol	Permission	Simple meaning
r	Read	Open or view the file. For a directory, list its names.
w	Write	Change a file. For a directory, create/delete entries (subject to other rules).
x	Execute	Run a file. For a directory, enter/traverse it.

Who gets permissions?

Class	Meaning
u	User / owner of the file
g	Group associated with the file
o	Others — everyone else
a	All three classes: user, group and others

Read a permission string

```
-rwxr-xr--  1 miko developers  1200 Oct 30 app.sh
```

Breakdown: first character - means regular file; rwx owner can read/write/execute; r-x group can read/execute; r-- others can only read.

chmod — change permissions

```
chmod u+x app.sh       # add execute for owner\nchmod g-w report.txt   # remove write from grou\nchmod o-r
```

Number method: r=4, w=2, x=1. Add the numbers for each class. 7=4+2+1=rwx; 5=4+1=r-x; 4=r--. Therefore 755 means owner rwx, group r-x, others r-x.

chown, chgrp, umask and sudo

```
sudo chown miko app.log             # change owner\nsudo chown miko:developers app.log  # owne and group\n
```

umask is a mask that removes permissions from the default permissions used when creating files and directories. A common umask of 022 usually results in files 644 and directories 755. Exact defaults can vary by application.

Special permission: sticky bit

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```
sudo chmod +t /shared
```

A sticky-bit directory (often shown as t at the end, such as drwxrwxrwt) lets users create files in a shared writable directory, but generally only the file owner, directory owner, or root can delete/rename a file. The classic example is /tmp.

> Captain Root:
> “Least privilege! Give people only the access they need — no more.”

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4. Processes, services and scheduling

A process is a running program. Every process has a Process ID (PID). A service is a background program that provides a function, such as a web server or database.

> Miko:
> “I opened a browser. Is that a process?”

> Kai:
> “Yes! The running browser is a process. The operating system tracks it using a PID.”

Inspect processes

```
ps                        # processes in this terminal\nps aux                    # detailed list of proces
```

Signals: ask a process to stop

Command / signal	Meaning
kill PID	Send the default SIGTERM request to a process.
kill -15 PID	SIGTERM: politely ask the process to stop.
kill -9 PID	SIGKILL: force stop; process cannot clean up.
pkill nginx	Send a signal to processes matching a name.

Best practice: Try a graceful stop first. Use SIGKILL only when necessary because the process cannot handle cleanup.

Foreground and background jobs

```
sleep 300                 # run in foreground\nsleep 300 &               # start in background\njobs
```

systemd and services

Many modern Linux distributions use systemd as the service manager. Use systemctl to inspect and control services.

```
sudo systemctl status nginx    # check status\nsudo systemctl start nginx     # start service\nsudo systemct
```

Logs with journalctl

```
journalctl                # view system journal\njournalctl -u nginx         # logs for nginx\njournalctl
```

Scheduling with cron

Cron runs commands on a schedule. Edit your own crontab with crontab -e. Format: minute hour day-of-month month day-of-week command.

```
crontab -e\n# Run a script every day at 2:30 AM\n30 2 * * * /home/miko/backup.sh
```

Cron memory trick: minute → hour → day of month → month → day of week. Check the script path, permissions, environment and logs if a cron job fails.

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5. Pipes, redirection and log investigation

This chapter is your detective toolkit. Logs are like CCTV recordings for a server. You search them to understand what happened.

> Miko:
> “The website is down! Should I restart everything?”

> Kai:
> “Not yet. First collect evidence. Check service status, recent logs, disk space and the exact error.”

Redirection

Symbol	Meaning
>	Send standard output to a file; overwrite the file.
>>	Append standard output to a file.
2>	Send standard error to a file.
&>	Send standard output and standard error to a file (supported by bash).
<	Read standard input from a file.
|	Send output from one command into another command.

```
echo 'hello' > hello.txt       # overwrite/create\necho 'again' >> hello.txt     # append\nls missing-file 2
```

Useful text-search commands

```
cat app.log                 # print file\nless app.log                # scroll through file\ntail app.log
```

Pipes combine commands

```
grep -i 'error' app.log | tail -n 20\n# Find errors, then show the latest 20 matching lines\n\nps aux | grep
```

Disk and memory checks

```
df -h              # filesystem disk space\ndu -sh /var/log    # size of a directory\nfree -h                # mu
```

A calm troubleshooting sequence

Order	Check
1	Confirm the symptom: what is failing, since when, and who is affected?
2	Check service: systemctl status SERVICE
3	Check logs: journalctl -u SERVICE -n 50 --no-pager
4	Check resources: df -h, free -h, top
5	Check network/listening ports: ss -tulpn, curl, ping (where appropriate).
6	Make one safe, approved change; then verify and document.

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6. Linux networking essentials

A server needs networking to communicate with users, databases, APIs and other servers. Think of an IP address as a building address and a port as a room number for a particular service.

Miko:
“The website URL opens nowhere. Is Linux broken?”

Kai:
“Maybe DNS, routing, firewall, the service, or the application. We test one layer at a time.”

Key terms

Term	Simple meaning
IP address	The network address of a device.
Port	A numbered endpoint used by an application; for example, HTTPS commonly uses 443.
DNS	Translates names such as example.com into IP addresses.
Gateway	The next-hop router used to reach other networks.
Firewall	Rules that allow or block network traffic.
TCP	Connection-oriented transport protocol; reliable ordered delivery.
UDP	Connectionless transport protocol; often used where low overhead matters.

Useful commands

```
ip addr                # view network interfaces and IPs\nip route              # view routing table\nhost
```

When a web server is not responding

```
sudo systemctl status nginx\nsudo journalctl -u nginx -n 50 --no-pager\nsudo ss -tulpn | grep ':80'\ncurl -I
```

Interpretation examples

Observation	Possible meaning / next check
Service is inactive	Check why it stopped in service logs.
Nothing listens on port 80	Check service config and startup.
Local curl works, remote fails	Check security group, firewall, routing, load balancer and DNS.
DNS name fails but IP works	Investigate DNS resolution.
Disk is 100% full	Find large files/logs safely; follow cleanup/change process.

Important cloud note: On AWS, Linux troubleshooting may also involve Security Groups, Network ACLs, route tables, load balancers, target health and DNS records. A Linux command cannot fix an AWS network rule by itself.

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7. Mini production drama: the website is down!

Scene: It is 10:15 AM. A user reports that the company website is not opening. You are the junior DevOps engineer on call.

> Alert Bot:
> “HTTP check failed. Website may be unavailable.”

> Miko:
> “I will restart the server immediately!”

> Kai:
> “Pause. A restart without evidence can hide the real cause. Let's investigate.”

Your mission: follow the clues

Step	Command	What you are checking
1	systemctl status nginx	Is the web service running?
2	journalctl -u nginx -n 50 --no-pager	What errors appeared recently?
3	ss -tulpn | grep ':80'	Is anything listening on port 80?
4	curl -I http://localhost	Does the app respond locally?
5	df -h	Is the disk full?
6	free -h; top	Is memory/CPU under pressure?

Sample clue: nginx is active, port 80 is listening, but the application returns 502 Bad Gateway.

> Miko:
> “The web server is alive, but it cannot get a good response from the backend. I should check the upstream application service, its port, logs and connectivity.”

That is a good answer. A 502 often means a gateway or proxy received an invalid response from an upstream server. It does not automatically mean nginx itself is down.

A simple interview answer

“First, I will understand the impact and confirm the issue. I will check the web service status and recent logs, verify whether the expected port is listening, and test the application locally using curl. Then I will check system resources and backend connectivity. I will make a controlled change only after identifying the likely cause, verify recovery, and communicate the update.”

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8. Quick revision and interview practice

Must-know commands

Task	Command
Where am I?	pwd
List hidden files	ls -la
Find current user	whoami
Show OS details	cat /etc/os-release
Show kernel version	uname -r
Find a file	find /path -name filename
Search text	grep 'word' file
Check disk	df -h
Check folder size	du -sh folder
Check memory	free -h
View processes	ps aux / top
Check service	systemctl status name
Read service logs	journalctl -u name
Check IP and route	ip addr / ip route
Check listening ports	ss -tulpn

Interview questions — answer in your own words

Question	Simple answer
What is Linux?	An open-source, Unix-like operating system family used on desktops, servers and cloud systems.
Kernel vs shell?	Kernel manages hardware/resources; shell interprets commands from the user.
What is a PID?	A unique process identifier assigned to a running process.
SIGTERM vs SIGKILL?	SIGTERM requests a graceful stop; SIGKILL forces termination.
What is chmod?	A command to change file or directory permissions.
What is chown?	A command to change ownership.
What is sudo?	A way for an authorized user to run a command with elevated privileges.
What is systemd?	A common init and service manager on modern Linux distributions.
What is cron?	A scheduler for recurring commands or scripts.
What does df -h show?	Filesystem disk space in human-readable units.
What does grep do?	Searches text for matching patterns.
What is a pipe?	It sends one command's output to another command's input.

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Tiny daily practice plan (30–40 minutes)

Time	Practice
5 min	Read one concept and explain it aloud.
10 min	Type 4–5 commands in a safe Linux lab.
10 min	Create a small task: files, permissions, process or logs.
5–10 min	Answer one interview question without reading.

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9. Safe hands-on lab

Run these in your own Linux VM, WSL environment or practice EC2 instance. Avoid experimenting on a production server or with important personal files.

Mission A — build a tiny folder world

```
mkdir -p ~/linux-drama/logs\ncd ~/linux-drama\ntouch notes.txt logs/app.log\necho 'INFO applica
```

Mission B — permissions

```
chmod 640 notes.txt\nls -l notes.txt\n# Expected permission pattern: -rw-r-----
```

Mission C — log detective

```
printf 'INFO started\\nERROR database timeout\\nINFO retry\\nERROR connection refused\\n' > log
```

Mission D — inspect your machine

```
whoami\nhostname\nuname -r\ncat /etc/os-release\ndf -h\nfree -h\nps aux | head
```

Self-check — can you explain these?

Check	Your goal
I can navigate	Use pwd, ls, cd and explain absolute vs relative paths.
I can manage files	Use mkdir, touch, cp, mv, cat and safe removal.
I understand access	Explain r/w/x, user/group/others and chmod 755 vs 644.
I can inspect processes	Find a PID, explain SIGTERM and check a service.
I can read logs	Use journalctl, tail and grep to locate errors.
I can troubleshoot basics	Check service, port, disk, memory and local HTTP response.

Kai:
“You do not become a Linux expert by memorizing 100 commands in one night. You become confident by understanding a few commands and using them correctly.”

You’ve got this. Small practice every day beats one exhausting study session.

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